

Q: If $|A| \neq 0$, then $|A^t| \neq 0$ and
 $(A^t)^{-1} = (A^{-1})^t$.

Sol: Since $|A| = |A^t|$

If $|A| \neq 0$, then $|A^t| \neq 0$

To prove $(A^t)^{-1} = (A^{-1})^t$

We have to show that

$$(A^t)(A^{-1})^t = I = (A^{-1})^t(A^t)$$

Now

$$\begin{aligned}(A^t)(A^{-1})^t &= (A^{-1}A)^t \quad (\text{as } (AB)^t = B^t A^t) \\ &= I^t \\ &= I\end{aligned}$$

$$\begin{aligned}\text{and } (A^{-1})^t(A^t) &= (AA^{-1})^t \quad (\text{as } (AB)^t = B^t A^t) \\ &= I^t \\ &= I\end{aligned}$$

Hence $(A^t)^{-1} = (A^{-1})^t$.

Q: Find area of the triangle with vertices
 $(3, 3), (4, 0), (-2, -1)$.

Sol: Given vertices are

$$(3, 3), (4, 0), (-2, -1)$$

Using

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

we get

$$\Delta = \frac{1}{2} \begin{vmatrix} 3 & 3 & 1 \\ 4 & 0 & 1 \\ -2 & -1 & 1 \end{vmatrix}$$

$$\begin{aligned}\Delta &= \frac{1}{2} [3(0+1) - 3(4+2) + 1(-4+0)] \\ &= \frac{1}{2} [3 - 24 - 4]\end{aligned}$$

$$\Delta = \frac{1}{2} |(3-28)|$$

$$= \frac{1}{2} |(-25)|$$

$$\Delta = \frac{25}{2} \text{ Square unit}$$

Q:- Show ~~$u = (-2, 3, 1, 4)$~~

Show $u = (-2, 3, 1, 4)$ and $v = (1, 2, 0, -1)$ are orthogonal vectors in $\mathbb{R}^4$. Also find angle between u and v

Sol:- Given

$$u = (-2, 3, 1, 4) \text{ and } v = (1, 2, 0, -1)$$

Now

$$\begin{aligned} \langle u, v \rangle &= \langle (-2, 3, 1, 4), (1, 2, 0, -1) \rangle \\ &= (-2)(1) + (3)(2) + (1)(0) + (4)(-1) \\ &= -2 + 6 + 0 - 4 \\ &= -6 + 6 \end{aligned}$$

$$\langle u, v \rangle = 0$$

Hence u and v are orthogonal vectors.

If θ is the angle between u and v then

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

$$\cos \theta = \frac{0}{\|u\| \|v\|}$$

$$\cos \theta = 0$$

$$\text{Thus } \theta = 90^\circ.$$